

Urinary metabolomics analysis reveals molecular linkages for bronchopulmonary dysplasia in preterm infants less than 32 weeks gestation

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Introduction

Bronchopulmonary dysplasia (BPD) is a chronic lung disease that primarily affects premature infants who require mechanical ventilation or oxygen therapy for respiratory distress syndrome. Despite there being a general improvement in lung function and a reduction in the symptoms associated with BPD within the first six months, most infants experience a significant reduction in symptoms by the age of two or three years.^{1, 2}

This study is aimed to identify the metabolic signatures of BPD severity related to gestational age (GA) using ¹H-NMR spectroscopy in the early life of preterm infants

Methods

Study population

Preterm infants less than 32 weeks gestation and full-term infants greater than 37 weeks gestation were prospectively enrolled into this study. BPD was diagnosed as requirement of oxygen supplementation either at 28 days postnatal age or 36 weeks postmenstrual age (PMA) based on the 2019 revision of National Institute of Child Health and Human Development (NICHD) criteria.³

Gestational age:

- GA > 37 weeks
- GA 28-32 weeks
- GA < 28 weeks groups

BPD severity:

- Healthy controls (HC)
- no or mild BPD (No+Mild BPD)
- Moderate or severe BPD (M+S BPD) groups.

NMR data processing and analysis

The raw ¹H-NMR spectra were imported into NMRProcFlow.⁴ To adjust the misalignment of NMR spectra alignment, parametric time warping and least-squares algorithm were performed. Using the method of variable size and intelligent bucketing, the spectra bucketing was performed and analyzed. Metabolites were identified by using the Chenomx NMR Suite 8.1 software (Chenomx Inc., Edmonton AB, Canada).

Results

Population characteristics

A total of 139 infants completed a 6-month follow-up period, of which 89 infants less than 32 weeks gestation, were enrolled into this study. The comparisons of the baseline characteristics between three groups categorized by different GA and BPD severity at the corrected age of 6 months are shown in Table 1.

Characteristics	GA			P-value	BPD severity			P-value
	>37 weeks (n=50)	28-32 weeks (n=48)	<28 weeks (n=41)		HC (n=50)	No+Mild (n=40)	M+S (n=49)	
Sex, male	20 (40.0%)	22 (45.8%)	20 (48.8%)	0.688	20 (40.0%)	19 (47.5%)	23 (46.9)	0.714
Gestational age (wk)	39.1 ± 0.8	30.7 ± 0.9	26.1 ± 1.4	<0.001	39.1 ± 0.80	30.2 ± 1.70	27.2 ± 2.40	<0.001
Birth body weight (g)	3171.3 ± 365.7	1369 ± 273.9	775.8 ± 203.7	<0.001	3171.3 ± 365.7	1404.9 ± 287.2	843.4 ± 241.8	<0.001
Age, corrected (month)	7.08 ± 1.29	7.65 ± 3.4	8.08 ± 2.07	0.144	7.08 ± 1.29	8.13 ± 3.71	7.62 ± 1.91	0.124
Body weight (g)	7.87 ± 0.75	7.71 ± 1.24	7.14 ± 1.42	0.009	7.87 ± 0.75	8.05 ± 1.00	6.96 ± 1.40	<0.001
Body height (cm)	67.31 ± 2.63	66.36 ± 3.42	65.53 ± 5.05	0.081	67.31 ± 2.63	67.50 ± 3.15	64.74 ± 4.64	<0.001
BMI (kg/m ²)	17.36 ± 1.25	17.44 ± 2.05	16.38 ± 1.55	0.005	17.36 ± 1.25	17.64 ± 1.60	16.39 ± 1.97	0.001
Breastfeeding ≥ 6 months	28 (56.0%)	15 (31.2%)	19 (46.3%)	0.046	28 (56.0%)	14 (35.0%)	20 (40.8%)	0.110
Sepsis, ever	1 (2.0%)	12 (25.0%)	30 (75.0%)	<0.001	1 (2.0%)	9 (22.5%)	33 (68.8%)	<0.001

Metabolites significantly associated with different GA and BPD severity

Nineteen and 17 urinary metabolites between groups in different GA and BPD severity selected with a *P*-value of < 0.05 are respectively showed in Figure 2C. A Venn diagram showed the distribution of the metabolites that associated with different GA and BPD severity (Figure 1A).

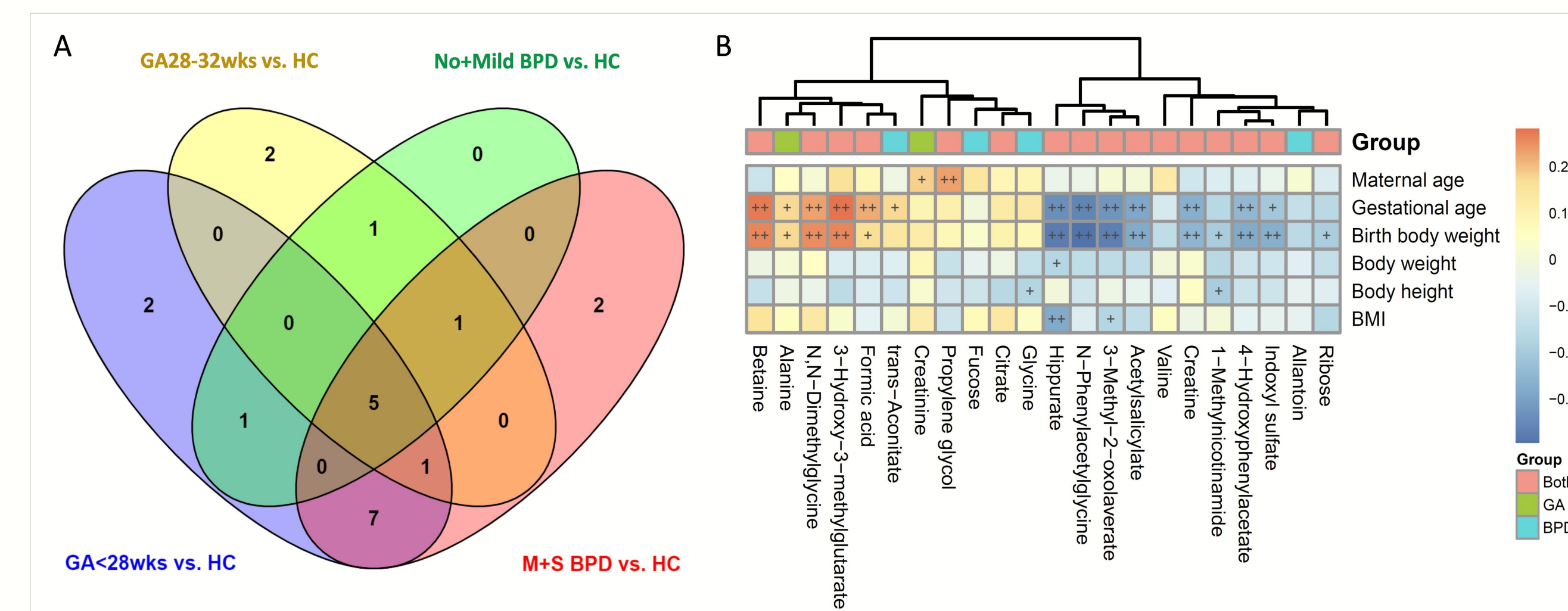


Figure 1. Venn diagram of the distribution of the metabolites significantly associated with different GA and BPD severity (A), and heatmap of Spearman's rank correlation coefficients between these metabolites and baseline characteristics (B). Total number of significantly differentially expressed metabolites in each set and in the overlapping areas is indicated. Acetylsalicylate, hippurate, 3-hydroxy-3-methylglutarate, N-phenylacetylglutamine, and creatine are in the common area. Color intensity represents the magnitude of correlation.

Five common metabolites including acetylsalicylate, hippurate, 3-hydroxy-3-methylglutarate, N-phenylacetylglutamine, and creatine were significantly associated with GA and BPD severity. Furthermore, metabolites related to GA and BPD severity were strongly associated with gestational age and birth body weight, but not body weight and BMI at the corrected age of 6 months (Figure 1B, *P* < 0.01).

Dynamic metabolic changes across different GA and BPD severity

C-means clustering was performed using R software and package *Mfuzz* and this resulted in 3 modules of metabolites that showed distinctive temporal patterns of GA (Figure 2A) and BPD severity (Figure 2B) characterized by their initial abundance of metabolites and variation in expression at the different status of groups. Compared to healthy infants, acetylsalicylate, hippurate, 3-methyl-2-oxovalerate, N-phenylacetylglutamine, indoxyl sulfate, 4-hydroxyphenylacetate, and creatine were significantly higher in infants with GA < 28 weeks and with moderate to severe BPD (*P* < 0.01). By contrast, 3-hydroxy-3-methylglutarate, N,N-dimethylglycine, and betaine were significantly lower in these infants (Figure 2C, *P* < 0.01).

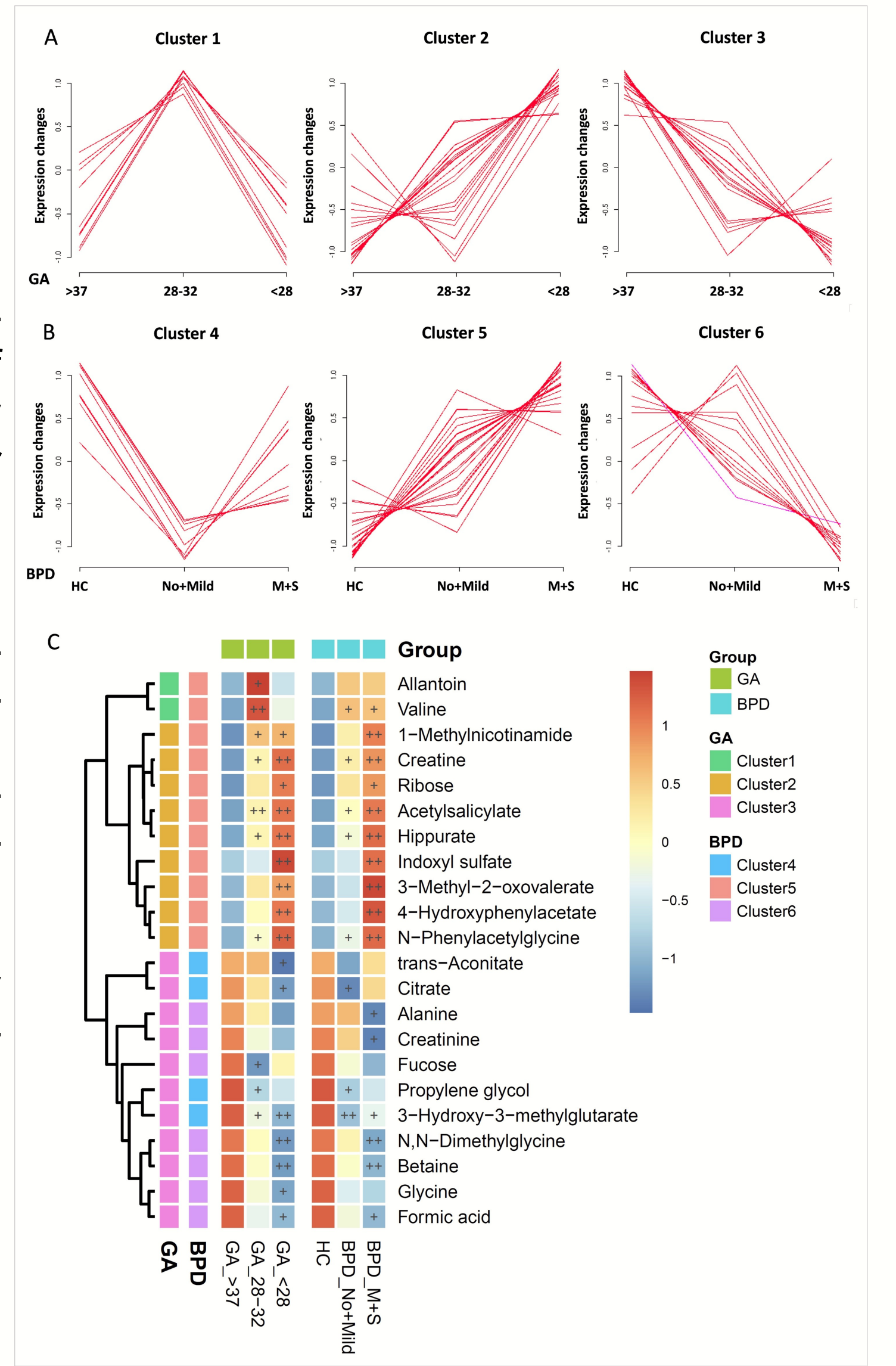


Figure 2. Dynamic metabolic changes across different GA (A) and BPD severity (B), and heatmap of significances of differentially expressed metabolites in comparison with healthy infants (C). Color intensity represents the integral regions of NMR signals of metabolites. Red color represents increase in abundance, blue color represents relative decrease. + symbol means a *P*-value < 0.05; ++ symbol means a *P*-value < 0.01. GA, gestational age; BPD, bronchopulmonary dysplasia; NMR, nuclear magnetic resonance.

Conclusion

Urinary metabolomic analysis provides the identification of a plausible association between perturbations in the metabolic pathways of amino acids and the risk of bronchopulmonary dysplasia (BPD) in preterm infants.

Reference

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